

# Indole-functionalized polythiourethane/tetrapodal shaped ZnO nanocomposites for eco-friendly marine biofouling control

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## ABSTRACT

Antifouling coatings for marine use are a widely employed method for controlling the settlement and growth of unwanted aquatic organisms on surfaces. In this study, indole derivative N-(1H-5-bromo indole-3-ylmethyl) acrylamide (BIAM)-functionalized polythiourethane (PTU)/tetrapodal shaped zinc oxide (t-ZnO) nanocomposites (PTU/BIAM/t-ZnO) were synthesized through a Friedel–Crafts Alkylation reaction, thiol-ene click reaction and polyaddition reaction. Nanocomposites' surface characteristics, such as surface free energy, chemical composition, and surface morphology were investigated here. The mechanical characteristics and antifouling properties of the coating materials were also considered. These nanocomposites exhibited not only superior mechanical properties, due to the presence of both PTU and t-ZnO, but also promising antifouling properties owing to the incorporation of BIAM on their surfaces. The adhesion strength of the nanocomposites increased from 0.26 MPa to 0.68 MPa due to the incorporation of t-ZnO particles. Benefiting from the PTU matrix, the tensile strength of nanocomposite can reach over 57 MPa. The PTU/BIAM/t-ZnO nanocomposites demonstrated improved algae resistance and marine antifouling performance compared to the reference PTU/t-ZnO nanocomposite without indole derivative. The PTU/BIAM/t-ZnO nanocomposites exhibit resistance to fouling organisms due to the synergistic effect of the easy-to-clean characteristic of the PTU/t-ZnO and the reduction of the polar component of the surface free energy resulting from the inclusion of indole derivative BIAM on the surface. This, in turn, reduces the interaction between fouling organisms and the material surface. Furthermore, BIAM interferes with the homeostasis of cellular calcium ions, thereby hindering the attachment of fouling organisms. Under the combined action of PTU, BIAM, and t-ZnO, the synergistic effect of good antifouling properties, environmental compatibility, good mechanical properties and strong adhesion to the substrate was achieved.

## 1. Introduction

Marine biofouling refers to the unwanted accumulation of microorganisms, plants, and animals on human-made structures that are submerged in seawater. Marine biofouling can have negative impacts on the utilization and exploitation of marine resources, thereby leading to economic costs and ecological issues. For example in shipping, biofouling can result in reduced sailing speed [1], increased fuel consumption [2], increased greenhouse gas emissions [3], and the

migration of non-native species [4]. In order to address these problems, researchers have investigated various techniques for biofouling control, including mechanical cleaning [5], electrochemical approaches [6], and the application of antifouling coatings [7,8]. Despite these efforts, an ideal solution to biofouling control has yet to be achieved, due to the complexity of the marine environment and the diversity of fouling organisms.

Marine antifouling coatings are currently regarded as the most cost-effective and efficient method for preventing the settlement of fouling

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organisms on submerged surfaces [9]. Historically, tributyltin (TBT)-based antifouling coatings have been developed as an effective approach to preventing biofouling [10]. However, these coatings have been found to exert severe negative effects on the marine environment, including shell deformations, growth impediments, reproductive failure in oyster cultures, and interference with the endocrine systems of mollusks [11,12]. As a result, the International Maritime Organization (IMO) has globally banned the use of TBT-based coatings since 2008 in order to protect the marine environment and promote sustainable development [13]. Commercial antifouling coatings presently incorporate a substantial concentration of copper compounds as a substitute biocide for TBT [14]. While copper-containing coatings exhibit less toxic, the accumulation of copper in the marine environment may cause ecological problems [15]. Therefore, copper-containing coatings may potentially be prohibited in the future.

The trend in the development of biofouling control coatings is towards the creation of environmentally friendly materials that are effective at preventing the settlement of fouling organisms. Currently, there has been significant interest in developing coatings based on surface wettability, self-renewable coatings, and biomimetic coatings, thereby circumventing the dependence on hazardous biocidal agents [16–19]. Among these, fouling-release coatings based on low surface free energy and micro-nano structure are a promising alternative to conventional biocide-containing coatings because of their ability to prevent fouling organisms from adhering or to facilitate their removal after attachment. This is achieved through the low surface energy and low elastic modulus of these coatings. The incorporation of the nanoparticles into the polymer matrix significantly increases the hydrophobicity of the nanocomposites, leading to a micro-nano superhydrophobic surface structure [20]. By incorporating various nanoparticles (e.g., alumina nanorod, spherical ZnO, Cu<sub>2</sub>O nanocube, etc.) into matrix, Selim et al. developed micro-nano superhydrophobic structure material, which showed improved fouling-release properties [21–26]. However, to achieve the fouling-release properties, a certain amount of shear force, e.g., through seawater washing or ship movement, is necessary [27]. Furthermore, marine coatings are often exposed to harsh environments and it is crucial that they possess superior mechanical properties and a strong adhesion to the substrate to ensure their longevity and effectiveness.

There are various approaches that have been used to improve the adhesion strength and mechanical performance of antifouling coatings, including the incorporation of nanoparticles [20] and chemical modification [28]. However, the surface properties of the nanoparticles can affect the agglomeration of the particles, which adversely affects the properties of nanocomposites. Tetrapodal shaped ZnO (t-ZnO) micro-nano particles have been shown to enhance the adhesion of composite coatings to the substrate attributed to their unique 3D morphology [29]. Additionally, the 3D structure of t-ZnO reduces the agglomeration of particles, thus the t-ZnO particles are more homogeneously dispersed inside the coating, thereby improving the physical properties of materials [30]. Furthermore, studies have demonstrated the absence of Zn ion release from the PTU/t-ZnO composite surface, making t-ZnO particles a safe option for use in a marine coating [31]. Polythiourethane (PTU) has several excellent properties that make it suitable for use as a matrix polymer in antifouling coatings, including environmental friendliness, high tensile strength, and high adhesion strength to substrates [32,33].

Preventing the colonization of bacteria and microalgae during the early stages of biofouling formation is essential in order to effectively control this process [8]. The utilization of antifouling compounds proves to be a successful approach [34]. Our previous research has demonstrated the inhibitory effects of indole derivatives on diatom growth [35]. Non-invasive micro-test technology suggested that marine algae are inhibited by the disruption of the Ca<sup>2+</sup> balance in their cells, which occurs when indole derivatives are present and leads to an increase in intracellular Ca<sup>2+</sup> efflux. Additionally, indole derivatives as the

widespread natural products in nature have been demonstrated to be environmentally friendly. Wang et al. studied the inhibitory effects of diindol-3-ylmethanes against larval settlement of *B. amphitrite*. Diindol-3-ylmethanes could be considered as low-toxicity antifoulants since their LC<sub>50</sub>/EC<sub>50</sub> ratios were larger than 15 [36]. In summary, indole derivatives have been shown to be effective at reinforcing the antifouling properties of coatings while having both high bioactivity and environmental friendliness [18].

In this study, to optimize the antifouling performance of PTU coatings, indole derivatives with carbon-carbon double bonds were incorporated into a PTU matrix through a thiol-ene click reaction, resulting in the creation of indole-functionalized PTU. T-ZnO micro-nano particles were incorporated into the modified PTU to create indole-functionalized PTU/t-ZnO nanocomposites (PTU/BIAM/t-ZnO). The solvent-free preparation process is environmentally friendly as it does not generate volatile organic compounds. The adhesion strength of the nanocomposite was doubled due to the incorporation of 5 wt% t-ZnO. The addition of BIAM had no negative effect on mechanical properties. The nanocomposites were further characterized for surface free energy, anti-algal activity, and antifouling performance in a marine field test. The combination of PTU, indole derivatives, and t-ZnO resulted in a significant enhancement of both biofouling control, providing a new approach for the development of environmentally friendly biofouling control coatings.

## 2. Materials and methods

### 2.1. Materials

5-Bromo indole, *N*-methylol acrylamide, and diiodomethane were obtained from Shanghai Aladdin Bio-Chem Technology Co., Ltd. (Shanghai, China). Absolute ethanol and sulfuric acid were provided by Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). 1,1'-azobis (cyclohexane-1-carbonitrile) was purchased from Merck KGaA (Darmstadt, Germany). Tetrafunctional pentaerythritoltetrakis (PETMP) and 1,6-diisocyanatohexane (HDI) were obtained from Performance Chemicals Handels GmbH (Buchholz in der Nordheide, Germany). The tetrapodal shaped micro-nano ZnO particles (t-ZnO) with a mean arm length of 50 μm were produced by the flame transport synthesis approach [37] at Phi-Stone AG (Kiel, Germany). All reagents were used as received.

### 2.2. Sample preparation

#### 2.2.1. Synthesis of *N*-(1*H*-5-bromo indole-3-ylmethyl) acrylamide (BIAM)

Indole derivative BIAM was synthesized via the Friedel-Crafts alkylation reaction and the detail process is as follows. A stirred solution of 5-bromo indole (9.80 g, 0.05 mol) and *N*-methylol acrylamide (6.10 g, 0.06 mol) in absolute ethanol (25 mL) was gradually treated with a dropwise addition of a suspension of sulfuric acid (5 mL) in absolute ethanol (10 mL) while being maintained in an ice bath. After 2 h of stirring in the ice bath, the mixture was gradually warmed up to room temperature and stirred for another 48 h at room temperature. The crude products were obtained after filtration, washing with deionized water to reach a pH of 7. The crude products were purified by recrystallization from ethyl alcohol to afford *N*-(1*H*-5-bromo indole-3-ylmethyl) acrylamide (BIAM). IR (KBr)  $\nu$ : 1455, 1532, 1594, 1647, 2900, 3263, 3315 cm<sup>-1</sup>; <sup>1</sup>H NMR (DMSO-*d*<sub>6</sub>, 600 MHz)  $\delta$ : 4.45 (d, 2H, CH<sub>2</sub>), 5.58 (m, 1H, CH<sub>2</sub>=C), 6.14 (m, 1H, CH<sub>2</sub>=C), 6.21 (m, 1H, CH=C), 7.19 (m, 1H, PyH), 7.34 (m, 1H, PhH), 7.34 (m, 1H, PhH), 7.74 (d, 1H, PhH), 8.40 (t, 1H, CH<sub>2</sub>NH), 11.16 (s, 1H, NH); <sup>13</sup>C NMR (DMSO-*d*<sub>6</sub>, 150 MHz)  $\delta$ : 34.11, 111.74, 112.58, 113.96, 121.54, 124.10, 125.57, 126.16, 128.83, 132.30, 135.47, 164.75; MS: C<sub>12</sub>H<sub>12</sub>BrN<sub>2</sub>O, meas.  $m/z$  = 279.0131,  $m/z$  = 279.0128; C<sub>12</sub>H<sub>11</sub>BrN<sub>2</sub>NaO, meas.  $m/z$  = 300.9950,  $m/z$  = 300.9947, the values of err [ppm] and mean err [ppm] are all -1.1.

The structure of N-(1H-5-bromo indole-3-ylmethyl) acrylamide (BIAM) is shown in Fig. S1. IR,  $^1\text{H}$  NMR,  $^{13}\text{C}$  NMR and MS spectra of BIAM are shown in the supporting information (Figs. S2–S5).

### 2.2.2. Synthesis of PTU/BIAM/t-ZnO

For the nanocomposite coating synthesis, the process was divided into two steps. The prepolymer (PETMP/BIAM) was first synthesized via thiol-ene click reaction between PETMP and BIAM [38]. PETMP and BIAM were added into a glass beaker. Subsequently, the reaction mixture was gradually warmed to  $85^\circ\text{C}$  and stirred for 0.5 h. 1,1'-azobis (cyclohexane-1-carbonitrile) as an initiator was added into the above system. The reaction was carried out at  $85^\circ\text{C}$  for 3 h.

Second, the prepolymer was naturally cool to room temperature naturally. T-ZnO particles were then added into the prepolymer and subjected to ultrasonic treatment for a duration of 15 min. The HDI component was subsequently added to the above mixture and stirred by hand. The resulting mixture was cast into a silicone mold and cured at  $85^\circ\text{C}$  for 24 h. The compositions of the coatings (PTU/BIAM/t-ZnO) are shown in Table S1. In order to avoid adverse effects on mechanical properties, the degassing step was required during the dispersion process of t-ZnO and HDI component. 5 wt% t-ZnO was added to the prepared copolymers due to the strong mechanical reinforcement and promising antifouling property, as supported by previous study [29]. The synthetic route of PTU/BIAM/t-ZnO is shown in Fig. 1.

## 2.3. Methods

### 2.3.1. Equipments

This study used the following equipments for the investigation: MLS-3750 autoclave sterilizer (Sanyo, Japan), MLR-350H Versatile Environmental Test Chamber (Sanyo, Japan), HZQ-F160 incubator shaker

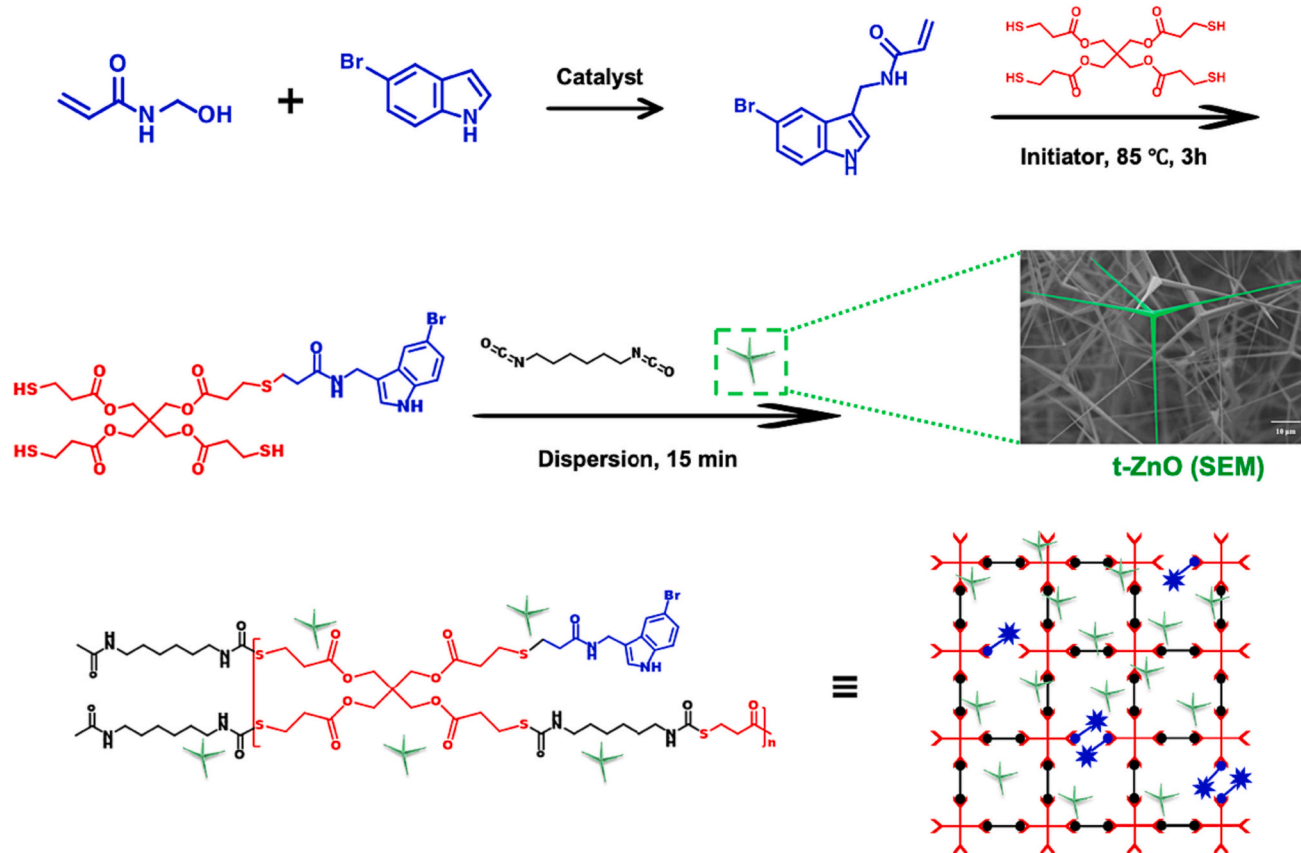
(Harbin Donglian, China), TU-1901 ultraviolet-visible spectrophotometer (Persee, China), Zwick 1445 universal testing machine (Zwick, Germany). IR spectra were recorded for KBr disc on a Tensor 27 Fourier Transform Infrared Spectrometer (Bruker, Germany).  $^1\text{H}$  NMR and  $^{13}\text{C}$  NMR spectra were recorded with AVANCE III (600 MHz) instrument (Bruker, Switzerland) in DMSO- $d_6$  as the solvent, using TMS as an internal standard, and chemical shifts were indicated in parts per million ( $\delta$ ). The surface morphology was determined using SIGMA 500 (Zeiss, Germany). Contact angles were measured by a DSA100 drop shape analyser (KRÜSS, Germany).

### 2.3.2. Minimum inhibitory concentration (MIC) test of BIAM

The minimum inhibition concentration was determined by a two-fold serial dilution test [39,40]. The bacterial strains *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) were incubated in nutrient broth medium to exponential phase. The bacterial suspensions were diluted to a concentration of  $10^6$  cfu/mL. BIAM were diluted successively to 1, 0.5, 0.25, 0.12, 0.06, 0.03 mg/mL with a two-fold dilution. The BIAM dilutions (2 mL) and bacterial suspensions (200  $\mu\text{L}$ ) were mixed with a vortex mixer and incubated at  $37^\circ$  for 24 h in an incubator shaker. The bacterial growth status was determined by measuring the light absorbance at 600 nm. Bacterial suspension with chlorothalonil (a broad-spectrum fungicide) was used as a positive control [41,42], and the one without any compound was used as a negative control.

### 2.3.3. Algal growth inhibition assay of BIAM

*Skeletonema costatum* (*S. costatum*), a globally distributed diatom species inhabiting tropical and polar seas, is commonly employed as a test species [43,44]. The algal solutions in the exponential growth phase were serially diluted and their concentrations were calculated using a cell counting plate. The standard curve was established according to the



**Fig. 1.** Synthetic route of PTU/BIAM/t-ZnO: (1) Synthesis of BIAM via Friedel–Crafts Alkylation reaction; (2) Synthesis of prepolymer (PETMP/BIAM) via thiol-ene click reaction; (3) Synthesis of PTU/BIAM/t-ZnO via polyaddition reaction.

various concentrations of algal solutions and the corresponding light absorbances at 680 nm. The unknown concentration of algal solution was determined by application of the standard curve.

EC<sub>50</sub> value (effective concentration for a 50 % growth inhibition) was determined according to the literature with slight modification as described below [45]. The initial concentration of algae solution was diluted to around 10<sup>5</sup> algal cells per milliliter. Acetone was chosen as a cosolvent to enhance the solubility of BIAM in water and served as a control group. The prepared algal solutions were exposed to the various concentrations of BIAM (0, 1, 2, 4, 8, 16 mg/mL). The samples were incubated at a light intensity of 4000 lx and 20 °C with a 12:12 h light:dark cycle. The light absorbances of samples were measured after three days, and the corresponding concentrations were calculated by application of the standard curve. Algal growth inhibition rates were calculated by the following formula:

$$S = \frac{S_0 - S_i}{S_0} \times 100\% \quad (1)$$

where S is algae colonization rate, S<sub>0</sub> is the concentration of control group, and S<sub>i</sub> is the concentration of experimental group. EC<sub>50</sub> values were determined through probit regression analysis using SPSS software. Chlorothalonil was served as a positive control and subjected to the aforementioned treatment.

#### 2.3.4. Surface morphology and elemental characterization of PTU/BIAM/t-ZnO

The surface morphology and elemental characterization of PTU/BIAM/t-ZnO were performed by the scanning electron microscope (SEM) at an acceleration voltage of 18 keV, equipped with energy-dispersive X-ray spectroscopy (EDX).

#### 2.3.5. Adhesion to substrate test of PTU/BIAM/t-ZnO

The adhesion of the nanocomposite to the AlMg3 substrate was evaluated by the adhesion test according to DIN EN ISO 462. The nanocomposites were painted on the AlMg3 substrates (80 mm × 80 mm) by brushes. After curing, a cylindrical AlMg3 stamp (50 mm in height, 20 mm in diameter) was glued to the center of the prepared surfaces using ethyl-2-cyanoacrylate glue (Henkel AG & Co. KgaA, Dusseldorf, Germany) and dried at room temperature for 24 h. The coating around the stamp was cut by a knife to define the test area. The adhesion test was performed using a Zwick universal testing machine with a pre-load of 30 N at a constant speed of 1 mm/min. Five tests were conducted for each material variation.

#### 2.3.6. Tensile test of PTU/BIAM/t-ZnO

The tensile test was performed according to the standard ISO 527. The dog-bone-shaped samples (20 mm in length, 5 mm in width, and 1 mm in thickness) were prepared by casting using silicone molds. The tensile test was investigated using a Zwick universal testing machine with a pre-load of 5 N at a constant speed of 1 mm/min. Five tests were conducted for each material variation.

#### 2.3.7. Contact angle and surface energy measurements of PTU/BIAM/t-ZnO

The surface wettability and surface free energy of the nanocomposites were determined by measuring the contact angles at room temperature. Seven points were randomly selected on different regions of the sample surface. The surface energy (γ<sub>s</sub>) with dispersive (γ<sub>s</sub><sup>d</sup>) and polar (γ<sub>s</sub><sup>p</sup>) components were calculated by the Owens-Wendt-Rabel-Kaelble method [46]. Water and diiodomethane were used as the reference liquids.

#### 2.3.8. Algal colonization assay of PTU/BIAM/t-ZnO

*S. costatum* suspension in the exponential growth phase was used for the experiment and diluted to 10<sup>5</sup> algal cells per milliliter using f/2

medium. The prepared samples were immersed in the sterilized glass vessels with algal suspension. The vessel containing only algal suspension was used as a control group. Subsequently, the above suspension was incubated with a 12:12 h light:dark cycle at a light intensity of 4000 lx and 20 °C. After 3 days of incubation, each sample was gently washed with deionized water to remove the unfixed algae. The algal cells adhering to the sample surfaces were shed from surface under the ultrasonic treatment. After that, the absorbances of treated algal solutions were determined and the corresponding concentrations were calculated by application of the standard curve. Algae colonization inhibition rates were calculated by the following formula:

$$T = \frac{T_0 - T_i}{T_0} \times 100\% \quad (2)$$

where T is algae colonization rate, T<sub>0</sub> is the concentration of control group, and T<sub>i</sub> is the concentration of experimental group. The above experiment was replicated on three occasions, and the mean value was adopted as the final result.

#### 2.3.9. Marine field test

The nanocomposites were coated on the AlMg3 substrates (100 mm × 100 mm). Three samples of each material variation were prepared and fixed randomly on a PVC plate. The sample plate was immersed in the Baltic Sea (Kiel Harbour, Germany) at a depth of 1 m from January 2021 to October 2021. The PVC plate was regularly taken out from seawater, gently washed with seawater, and photographed to record the status of biofouling.

#### 2.3.10. Statistical analysis

Statistical analysis of experimental results was performed with one-way ANOVA (Tukey test). In all analyses, a P value below 0.05 is considered to be statistically significant.

### 3. Results and discussion

#### 3.1. MIC value of BIAM

To investigate the antibacterial performance of BIAM, MIC assay was conducted. As shown in Table S2, the MICs of BIAM and chlorothalonil towards *E. coli* were both 0.06 mg/mL while MICs of BIAM and chlorothalonil towards *S. aureus* were 0.06 and 0.12 mg/mL, respectively. The results indicate that BIAM exhibits superior antibacterial performance compared to the reference chlorothalonil. This can be attributed to the mechanism of action of BIAM, which involves interference with the homeostasis of cellular calcium ions. Consequently, the reduction in intracellular Ca<sup>2+</sup> abundance impedes the growth and development of bacteria [42].

#### 3.2. Algal growth inhibition assay of BIAM

To further evaluate the antifouling performance of BIAM towards algae, the algal growth inhibition assay of *S. costatum* was conducted by measuring the light absorbance of the algal solution. The standard curve of light absorbance versus *S. costatum* concentration is depicted in Fig. S6. The correlation coefficient of the corresponding linear equation was greater than 0.99, indicating that there was a significant positive correlation between absorbance and concentration of the algal solution. Therefore, the measured light absorbance of the algal solution can be converted into concentration using the standard curve.

Based on the algal growth inhibition rates and the corresponding concentration of algal solutions, the EC<sub>50</sub> value was calculated. Chlorothalonil with broad-spectrum bactericidal activity showed a lower EC<sub>50</sub> value (6.44 mg/L), whereas BIAM had an EC<sub>50</sub> value of 7.86 mg/L. The above EC<sub>50</sub> values against *S. costatum* fulfill the U.S. Navy Program standards of EC<sub>50</sub> values which should be lower than 25.0 mg/L [47].

The BIAM showed comparable efficacy against *S. costatum* as chlorothalonil. Nevertheless, studies have revealed that the primary degradation product of chlorothalonil is 4-hydroxy-chlorothalonil [48], which is more toxic and persistent in natural environments. The existence of 4-hydroxy-chlorothalonil may cause inhibition of chlorothalonil degradation due to its toxicity to microorganisms associated with degradation [49]. In contrast, indole derivatives are more environmentally friendly and can be degraded into carbon dioxide and water [50,51]. Overall, BIAM can be used as a promising antifoulant against the growth of algae.

### 3.3. IR and $^1\text{H}$ NMR spectra of PETMP/BIAM and PTU/BIAM

IR spectroscopy was performed to identify functional groups present in the nanocomposite coatings. Fig. 2a showed the IR spectra of the prepolymer PETMP/0BIAM, PETMP/1BIAM, PETMP/4BIAM and PETMP/7BIAM. The absorption peaks at 2948 and 2902  $\text{cm}^{-1}$  were assigned to the stretching vibration absorption of saturated hydrocarbons ( $\text{CH}_2$ ). The peak at 2565  $\text{cm}^{-1}$  was the typical SH stretching vibration. The stretching vibration absorption of the carbonyl group was characterized by an intense peak at 1729  $\text{cm}^{-1}$ . The peaks at 1247, 1190 and 1143  $\text{cm}^{-1}$  were assigned to the asymmetric and symmetric stretching vibration absorptions of  $\text{C}-\text{O}-\text{C}$ . The aromatic group at 1533  $\text{cm}^{-1}$  and the  $\text{N}-\text{H}$  bond at 3379  $\text{cm}^{-1}$  were present in the spectra of PETMP/1BIAM, PETMP/4BIAM and PETMP/7BIAM, but not in that of PETMP/0BIAM. With increasing mole fraction of indole derivative, the intensities of absorption peaks (aromatic group and  $\text{N}-\text{H}$  bond) increased gradually. Due to the small amount of indole derivative in PETMP/1BIAM, the corresponding peaks of aromatic radical and  $\text{N}-\text{H}$  bond were not clearly visible. The characterization indicated that indole derivative BIAM was copolymerized with PETMP.

The formation of prepolymer was further confirmed by the presence of characteristic peaks in the  $^1\text{H}$  NMR spectra (Fig. 2b). The peaks at 2.4 ppm in PETMP/0BIAM, PETMP/1BIAM, PETMP/4BIAM and PETMP/7BIAM were attributed to the thiol group. The characteristic peaks (NH group and aromatic ring) in BIAM were used to determine the introduction of BIAM to the prepolymer. The NH signals of pyrrole ring and acylamino group were detected at approximately 11.1 and 8.4 ppm in the spectra of PETMP/1BIAM, PETMP/4BIAM and PETMP/7BIAM. The similar changes were found in the aromatic zone (7–8 ppm), where indole ring was commonly found in the spectra of PETMP/1BIAM, PETMP/4BIAM and PETMP/7BIAM. However, the NH and aromatic hydrogen signals were absent in the spectrum of PETMP/0BIAM. Additionally, the peaks ranging from  $\delta$  5.5 to 6.5 ppm corresponding to carbon-carbon double bond were absent in the spectra of PETMP/1BIAM, PETMP/4BIAM and PETMP/7BIAM. This illustrated that the indole derivative BIAM was introduced to the prepolymer via a thiol-ene click reaction.

Fig. 2c showed the IR spectra of copolymer PTU/0BIAM, PTU/1BIAM, PTU/4BIAM and PTU/7BIAM. The complete disappearance of SH peak was observed in the IR spectra, which indicated PETMP was reacted completely with BIAM and HDI. Another spectral change was obviously seen in the regions at 1682  $\text{cm}^{-1}$  when comparing Fig. 2a and c. This change was related to the formation of urea bonds. The excess of HDI over the stoichiometric ratio reacted with the water of surrounding humidity to form urea bonds [52]. The peak at 1514  $\text{cm}^{-1}$  was assigned to amide II (mixed contribution of the  $\text{N}-\text{H}$  in-plane bending, the  $\text{C}-\text{N}$  stretching, and the  $\text{C}-\text{C}$  stretching vibrations) [53,54]. The bands associated with aromatic skeletal vibrations overlapped with amide bands [55]. Overall, the formation of the desired product was confirmed by the IR and  $^1\text{H}$  NMR spectra of the prepolymer and copolymer.

### 3.4. SEM and EDX of PTU/BIAM/t-ZnO

The surface morphology and qualitative analysis of nanocomposites were investigated by SEM and EDX, as shown in Fig. 3. SEM micrographs revealed that all the nanocomposite coatings had the smooth surfaces and no cracks or pinholes were observed on the coating surfaces. There was no obvious difference on surface morphology between PTU/0BIAM/t-ZnO and PTU/1BIAM/t-ZnO, PTU/4BIAM/t-ZnO, PTU/7BIAM/t-ZnO. The presence of bromine (Br) in the BIAM was employed to characterize the amount of BIAM present on the surface. An obvious appearance of Br signal on the surface was observed after the introduction of BIAM. The sulfur (S) signal was used to represent PTU. The elemental sulfur and bromine were uniformly distributed across the sample surface as shown in Fig. 3. It indicated that BIAM was homogeneously present on the nanocomposite surfaces. With an increase in the amount of BIAM incorporated into the nanocomposites, there was a significant increase in the Br signal on the sample surface.

The above structural characterization data, such as chemical structure, surface morphology and elemental composition, were in accordance with the expected structures. This provides a safeguard for the relationship analysis of properties and structure.

### 3.5. Contact angle and surface free energy of PTU/BIAM/t-ZnO

As shown in Fig. 4, the wettability of PTU/BIAM/t-ZnO nanocomposites was evaluated by measuring the contact angle and surface free energy. The corresponding raw data are shown in the supporting information (Table S3). The water contact angle of PTU/0BIAM/t-ZnO without BIAM was around 71°. When increasing the BIAM concentration from 1 mol% to 7 mol%, the contact angle increased slightly from around 73° to 80°. The surface free energy of PTU/0BIAM/t-ZnO without BIAM was around 38.9  $\text{mJ}/\text{m}^2$ . The surface free energy decreased to 37.7  $\text{mJ}/\text{m}^2$ , 35.9  $\text{mJ}/\text{m}^2$  and 33.6  $\text{mJ}/\text{m}^2$ , respectively, when adding the BIAM in 1 mol%, 4 mol%, 7 mol%. The addition of

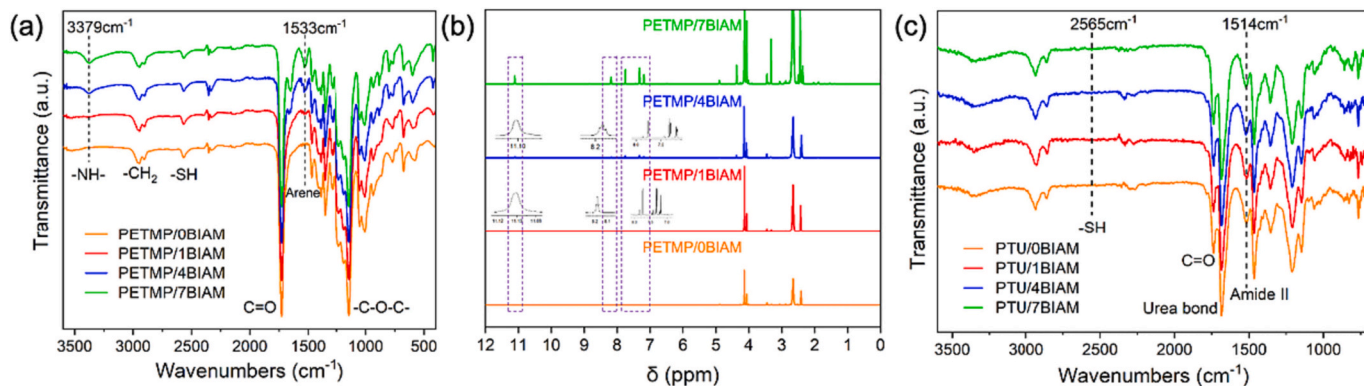
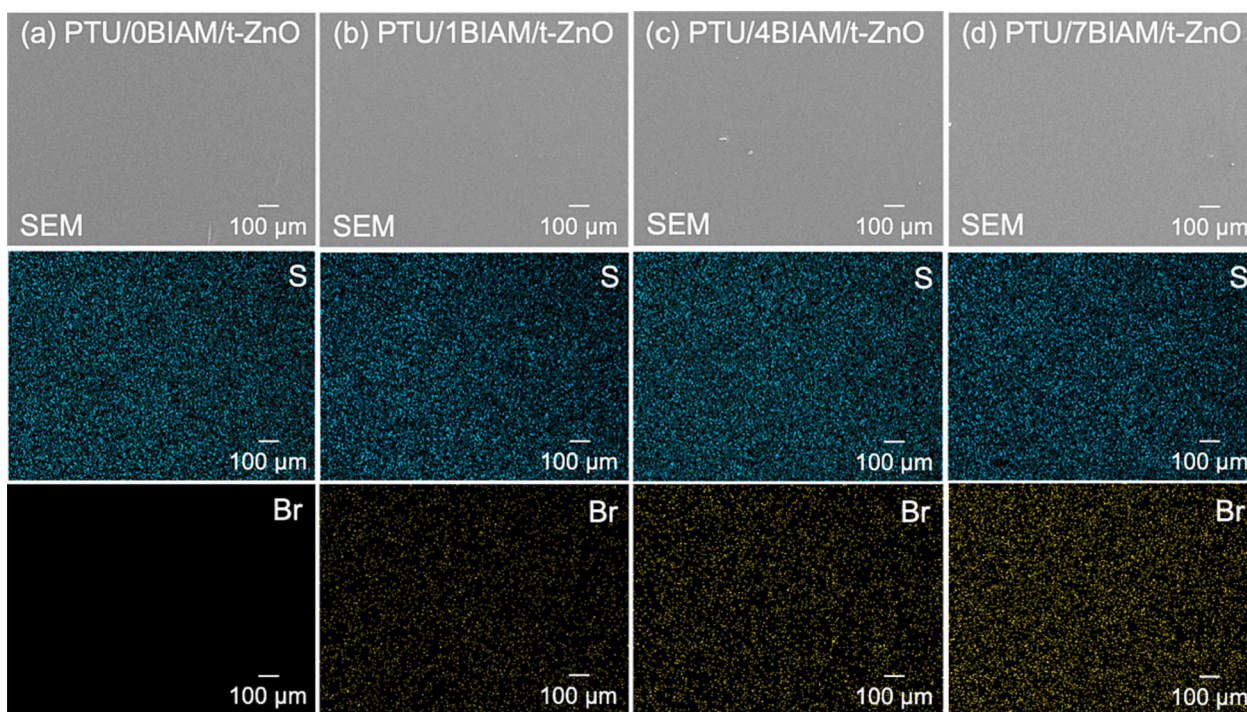
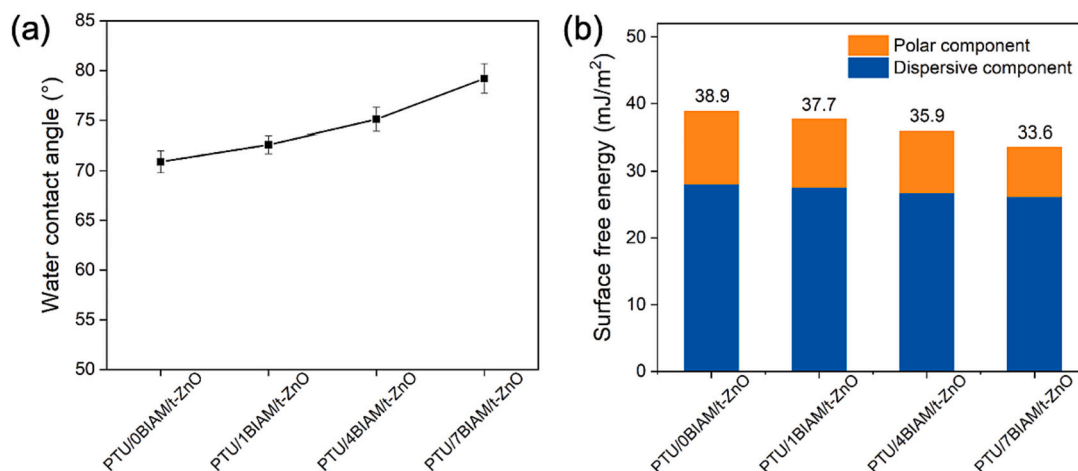


Fig. 2. (a) IR spectra and (b)  $^1\text{H}$  NMR spectra of PETMP/0BIAM, PETMP/1BIAM, PETMP/4BIAM and PETMP/7BIAM, (c) IR spectra of PTU/0BIAM, PTU/1BIAM, PTU/4BIAM and PTU/7BIAM.



**Fig. 3.** SEM and EDX images of (a) PTU/0BIAM/t-ZnO, (b) PTU/1BIAM/t-ZnO, (c) PTU/4BIAM/t-ZnO, and (d) PTU/7BIAM/t-ZnO. The Br signal in yellow is associated with BIAM, whereas the S signal in blue refer to PTU. The magnification in all images is identical. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



**Fig. 4.** Influence of various BIAM addition to PTU/BIAM/t-ZnO on contact angle (a) and surface free energy (b). Mean values with the corresponding standard deviations of water contact angle are shown.

hydrophobic indole derivative BIAM led to a reduction in both the dispersive and polar components of surface free energy, with the polar component showing a stronger decrease. This could contribute to reducing the interactions between fouling organisms and material surfaces by weakening the ability for strong polar interactions [56].

### 3.6. Adhesion strength of PTU/BIAM/t-ZnO to the substrate

The adhesion strength of the nanocomposites to the AlMg3 substrate is shown in Fig. 5a. The corresponding raw data are shown in the supporting information (Table S4). The adhesion strength of pure PTU was only about 0.26 MPa, while for all the nanocomposites with t-ZnO the adhesion strength was higher than 0.68 MPa. The enhanced adhesion strength for the nanocomposites with t-ZnO can be attributed to the

unique 3D shape of t-ZnO particles. The tetrapodal shape of the t-ZnO structure acted as micro hooks, which provided the mechanical interlocking between t-ZnO and substrates [29,57,58]. The mean values for adhesion range from 0.68 MPa to 0.77 MPa after introducing indole derivative BIAM to nanocomposites. There is no significant difference of adhesion strength to the substrate between PTU/1BIAM/t-ZnO, PTU/4BIAM/t-ZnO and PTU/7BIAM/t-ZnO. The introduction of indole derivative BIAM did not negatively impact the adhesion strength of the nanocomposite.

### 3.7. Tensile test of PTU/BIAM/t-ZnO

The tensile strength and elongation at fracture of the nanocomposites were characterized by tensile test. The corresponding raw data are

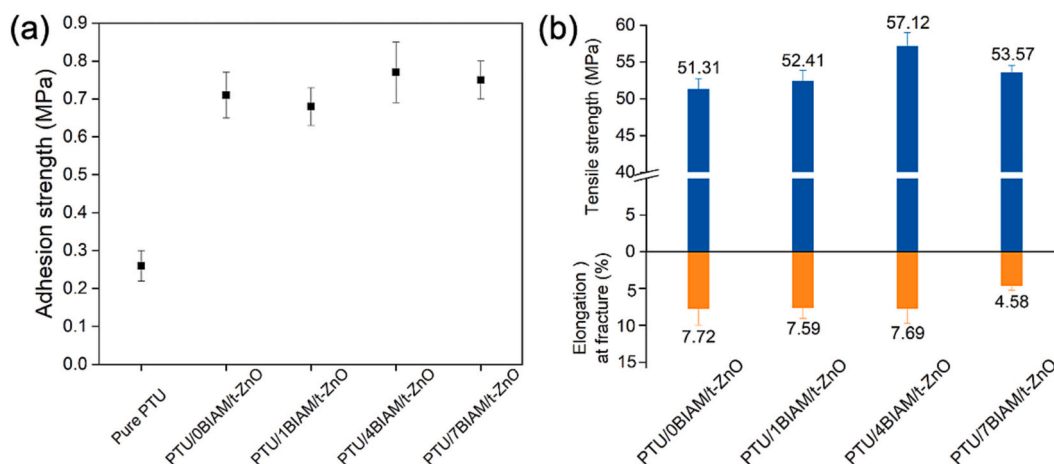


Fig. 5. (a) Adhesion strength to the substrate, (b) tensile strength, and elongation at fracture of PTU/BIAM/t-ZnO.

shown in the supporting information (Table S5). As shown in Fig. 5b, there was a significant difference between PTU/0BIAM/t-ZnO, PTU/4BIAM/t-ZnO and PTU/7BIAM/t-ZnO, while no significant difference between PTU/0BIAM/t-ZnO and PTU/1BIAM/t-ZnO could be observed for the tensile strength. The following order of tensile strength was observed: PTU/4BIAM/t-ZnO > PTU/7BIAM/t-ZnO > PTU/1BIAM/t-ZnO  $\approx$  PTU/0BIAM/t-ZnO. The tensile strength of PTU/4BIAM/t-ZnO showed a significant improvement of 11 % compared to that of PTU/0BIAM/t-ZnO. The increase was attributed to that the aromatic rings contained in the indole derivative BIAM can provide strong anti-rotational bonds which increase the rigidity of polymer chains [59–61]. The increased tensile strength of nanocomposite coatings benefited from the enhancement of chain rigidity. For example, due to the existing aromatic rings in polystyrene, the tensile strength of polystyrene is higher than that of polyvinyl [59–61]. For the tensile strength difference between PTU/0BIAM/t-ZnO and PTU/1BIAM/t-ZnO, a possible explanation is that only 1 mol% BIAM in PTU/1BIAM/t-ZnO had little effect on the change of anti-rotational bonds. The reduction of tensile strength when further increasing BIAM from 4 mol% to 7 mol% can be led by the change of cross-linked degree in copolymers. The incorporation of a substantial quantity of indole derivative reduced the crosslinking of PETMP and HDI, resulting in a high intermolecular slippage, thus decreasing the tensile strength [62,63]. Regarding the elongation at fracture, no significant distinction was observed for the nanocomposite without and with lower BIAM addition than 7 mol%. When adding BIAM in a concentration of 7 mol%, the elongation at fracture decreased sharply from 7.72 % of nanocomposite without BIAM to 4.58 %. This can be due to that the rigid indole rings reduce the flexibility of the polymer chain [63,64].

### 3.8. Algal inhibition test of PTU/BIAM/t-ZnO

Algal colonization usually occurs during the initial phase of marine biofouling which influences the whole biofouling process. Therefore, the antifouling performance of the samples against diatom *S. costatum* was evaluated in the laboratory (Fig. 6). The corresponding raw data are shown in the supporting information (Table S6). The inhibition rates of PTU/0BIAM/t-ZnO, PTU/1BIAM/t-ZnO, PTU/4BIAM/t-ZnO, and PTU/7BIAM/t-ZnO were 45 %, 51 %, 57 %, and 68 %, respectively. The nanocomposites with the addition of indole derivative BIAM showed good algal resistance due to their impact on the growth of algae by disrupting the calcium ions balance in algal cells. It was reported that halogenated indole derivatives could induce the efflux of cellular  $\text{Ca}^{2+}$ , which reduced the abundance of  $\text{Ca}^{2+}$  inside the algal cells. Thus, the normally stable physiological environment within the cell was disrupted [35,42]. Nakahata et al. reported that indole derivative 5, 6-dibromo-1,

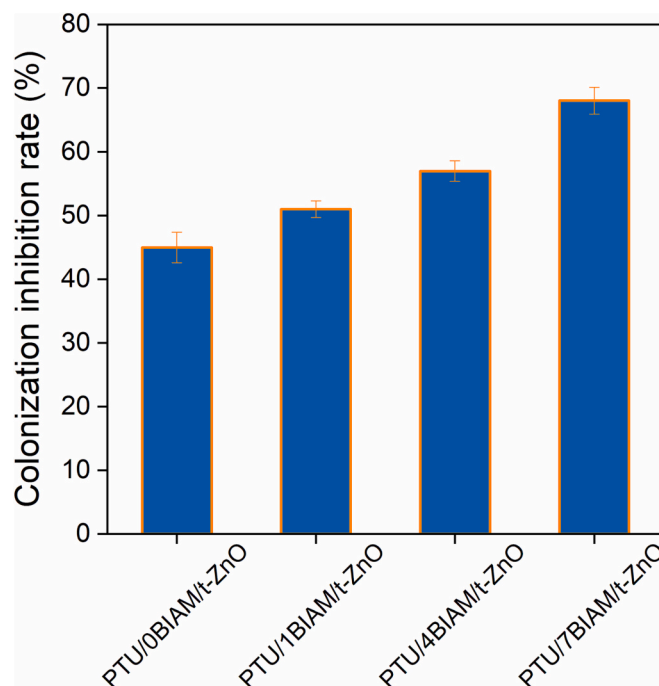


Fig. 6. Colonization inhibition rate of PTU/BIAM/t-ZnO against *S. costatum*.

2-dimethylgramine evoked  $\text{Ca}^{2+}$  release from skeletal muscle sarcoplasmic reticulum through ryanodine receptors [65]. Indole derivatives inhibited the normal operation of transmembrane transport, so indole derivatives were considered to act as an inhibitor of transmembrane transport of cellular calcium ions according to the literature [42]. Based on the EDX results of PTU/BIAM/t-ZnO, the greatest amount of indole derivative BIAM appeared on the surface of PTU/7BIAM/t-ZnO. Therefore, PTU/7BIAM/t-ZnO showed the greatest adverse impact on the physiological activity of *S. costatum*. On the other hand, the polar component of surface free energy was significantly decreased due to the introduction of BIAM to nanocomposite surface, which reduced the interactions between *S. costatum* and the material surface.

### 3.9. Marine field test

To further evaluate the antifouling property of PTU/BIAM/t-ZnO, the marine field test was conducted (Fig. 7). The reference AlMg3 substrate was eroded by seawater after one month. All coated surfaces were

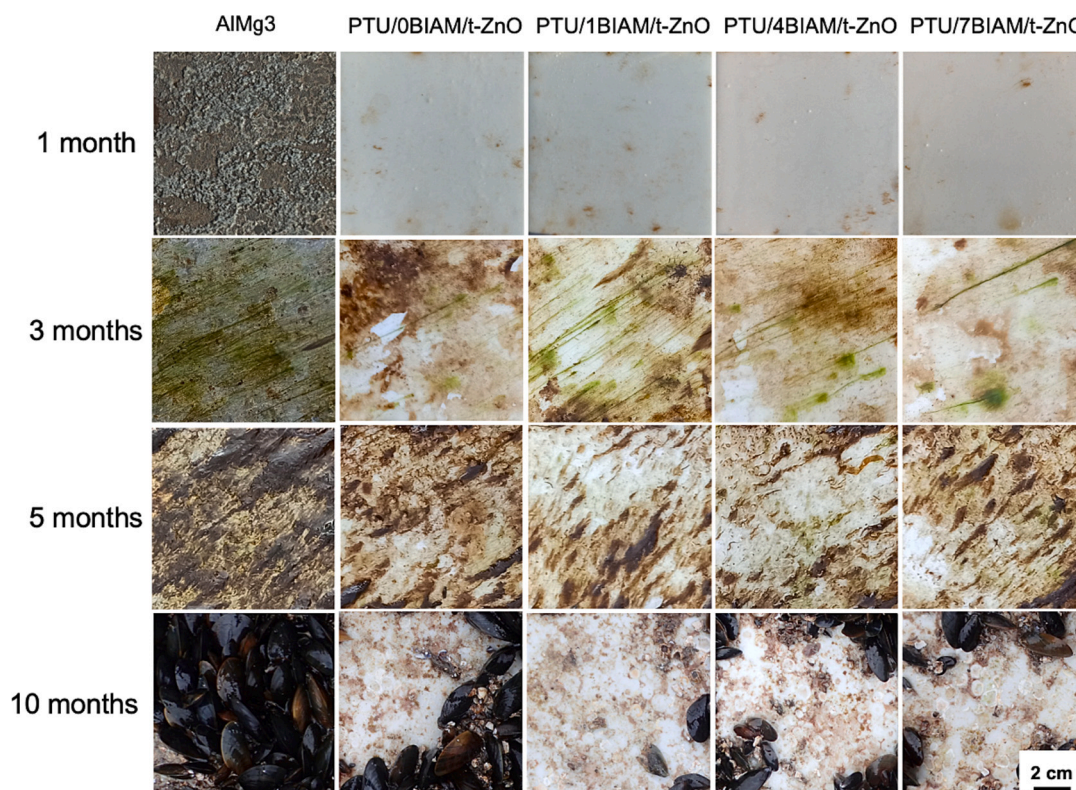


Fig. 7. Images of the surfaces of material variations (AlMg3, PTU/0BIAM/t-ZnO, PTU/1BIAM/t-ZnO, PTU/4BIAM/t-ZnO and PTU/7BIAM/t-ZnO) immersed in the sea for different periods. The magnification in all images is identical.

intact without damage or flaking, where only a small amount of biofilm adhered to the surfaces of the coatings. After three months of immersion, erosion occurred more intensively on the AlMg3 surface, accompanied by a significant accumulation of green algae. Compared to AlMg3, the coated surfaces were attached with fewer algae, especially PTU/4BIAM/t-ZnO and PTU/7BIAM/t-ZnO. The situation of biofouling in the fifth month was generally similar to that in the third month. Ten months after immersion, the AlMg3 surface was fully covered by large numbers of mussels while all the coating variations had less mussel coverage, especially those with BIAM addition. The few mussels presented on the corners of the samples with BIAM addition were due to the edge effect. Furthermore, no evidence of coating detachment was detected on any of the sample surfaces after 10 months of immersion in seawater, indicating their excellent adhesion strength to substrate.

### 3.10. Antifouling mechanism

Based on the results presented above, the antifouling mechanism of PTU/BIAM/t-ZnO was summarized in Fig. 8. The good antifouling properties of the nanocomposites are ascribed to the combined effect of PTU/t-ZnO with easy-to-clean characteristics [29], and indole derivative BIAM on the surface, in which the introduction of BIAM mainly reduces the polar component of surface free energy and BIAM also interferes with the homeostasis of cellular calcium ions to inhibit the attachment of fouling organisms. Specifically, the decrease of polar component of surface energy can be effective to reduce interactions between organisms and nanocomposite surface. Even though fouling organisms can adhere to the nanocomposite surface, attached organisms are easily removed. Furthermore, indole derivative BIAM can act on transmembrane transport of cellular calcium ions and trigger  $\text{Ca}^{2+}$  efflux from internal cell environment. The presence of BIAM can disrupt the homeostasis of  $\text{Ca}^{2+}$  inside the cells and cause detrimental effects on growth and development, which in turn leads to restraining the attachment of fouling organisms.

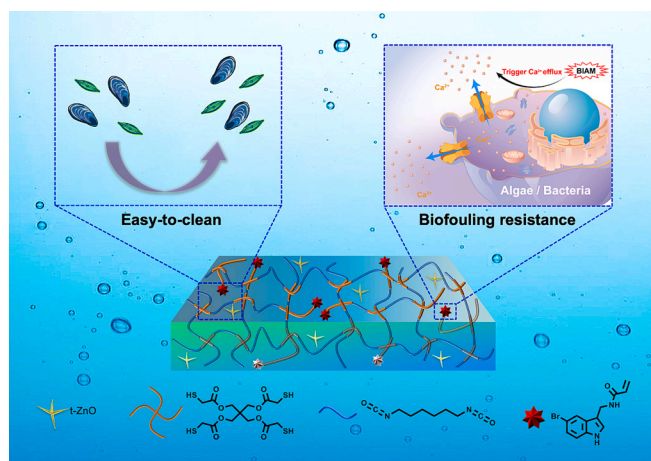


Fig. 8. Antifouling mechanism diagram of PTU/BIAM/t-ZnO. Part of the sketch was reproduced from Figdraw (Zhejiang, China) with permission.

### 4. Conclusions

A novel nanocomposite material, comprising of indole-functionalized polythiourethane (PTU) and tetrapodal shaped ZnO (t-ZnO), was synthesized in order to achieve a combination of good anti-fouling properties, environmental compatibility, good mechanical properties and strong adhesion to the substrate. The anti-bacterial and anti-algal properties of the indole derivative, BIAM, were evaluated and it was found that the halogenated indole derivative can reduce cellular calcium ion levels and inhibit the growth and development of micro-organisms. EDX analysis revealed that the indole derivative is homogeneously distributed on the surfaces of PTU/BIAM/t-ZnO. The improvement in adhesion strength was attributed to the unique

structure of the t-ZnO micro-nano particles. Benefiting from the addition of BIAM, PTU/BIAM/t-ZnO nanocomposites exhibit outstanding anti-algal and antifouling properties. Overall, the presented approach has shown promising biofouling control properties with great potential as an environmentally friendly marine coating.

### CRediT authorship contribution statement

**Kang Feng:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Haoyi Qiu:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Anna Gapeeva:** Conceptualization, Writing – review & editing. **Xia Li:** Conceptualization, Resources, Writing – review & editing, Supervision, Funding acquisition. **Yan Li:** Conceptualization, Writing – review & editing. **Sören Kaps:** Conceptualization, Writing – review & editing. **Yogendra Kumar Mishra:** Conceptualization, Resources, Writing – review & editing, Supervision, Funding acquisition. **Rainer Adelung:** Conceptualization, Resources, Writing – review & editing, Supervision, Funding acquisition. **Martina Baum:** Conceptualization, Resources, Writing – review & editing, Supervision, Funding acquisition. **Liangmin Yu:** Conceptualization, Writing – review & editing.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

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### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2023.107939>.

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